

Muhibu KANEKA

Geo-environmental and spatial data Analyst

PROFILE

Geo-environmental Spatial data analyst with a background in geology and a strong interest in the mining and natural-resources sector. Experienced in geospatial analysis, spatial data management, GIS, remote sensing and workflow automation, supporting multidisciplinary environmental related projects across office and field settings.

CONTACT

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LinkedIn

Webpage

TECHNICAL SKILLS

- ArcGIS Pro, QGIS, ENVI, Geo, Datamine
- PostgreSQL, PostGIS, SQL
- Geoserver, OpenLayers, Leaflet
- Python, HTML, CSS, JS
- Git (Github/Gitlab)
- PowerBI

CORE COMPETENCIES

- Spatial Database Development & Management
- Spatial Analysis (Qualitative & Quantitative)
- Remote Sensing & Satellite Image Processing
- Field Data Collection & Ground Truthing
- Data Quality Control & Validation
- Web mapping & Data Visualization

SOFT SKILLS

- Aadaptability
- Detail oriented
- Quick learner
- Strong communication skills

LANGUAGES

- Kiswahili : Native
- English : Fluent
- French : Fluent

HOBBIES

- Cycling
- Meditation
- Theatre
- Safari

Rreferee

Oswald Bomani

Hydrogeologist, Barrick

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EXPERIENCES

Geospatial analyst

INRAE, Aix-en-Provence | oct 2021 - current

Context :

- Development of an interactive web mapping application.

Key Responsibilities:

- Designed and administered spatial and non-spatial databases (PostgreSQL/PostGIS), ensuring data integrity and consistency.
- Extracted, cleaned, transformed, and loaded data from multiple sources (IGN, DFCI, CRIGE, BDIFF) into internal spatial databases.
- Built and maintained a Web GIS application to visualize spatial data and support stakeholder decision-making.
- Collaborated closely with multidisciplinary teams, defined data needs, delivering GIS outputs, and support project objectives.
- Prepared technical reports, GIS documentation, and user manuals for project teams and partners.

Geospatial analyst

INRAE, Aix-en-Provence | Mar - Sept 2021

Context :

- The development of an automated satellite image processing system for forest fire mapping and monitoring

Key Responsibilities:

- Improved a python-based workflows which automate satellite image processing, spatial analysis, and fire-mapping outputs.
- Developed customized workflows and automated processes to improve efficiency in data processing and visualization.
- Performed manual analysis of satellite images using GIS and remote sensing techniques, mirroring the automated workflow to test and validate its accuracy.
- Created an automation script for raster-to-vector format conversion and integrated the results into a spatial database.
- Conducted ground truthing for the verification of the efficiency of the automation tool

Remote sensing analyst

ESPACE (UMR 7300), Nice | Mar - July 2020

Context:

- The analysis of runoff due to suburban development on the “Brague catchment” (southern France)

Key Responsibilities:

- Processed satellite imagery to generate land cover and land-use maps
- Conducted spatial and statistical analyses to assess the impact of suburban development on runoff dynamics.
- Developed hydrological models incorporating runoff coefficients, precipitation, and infiltration parameters.
- Produced maps, analytical reports, and visual outputs to support environmental risk assessment.

Geologist (internship)

TPDC, Tanzania | Jan - Apr 2019

Context:

- Gravity prospecting for structural studies using remote sensing data in the southern Tanzania petroleum basins

Key Responsibilities:

- Conducted satellite gravity data acquisition, processing, and interpretation for structural analysis.
- Performed structural mapping (faults, folds, lineaments) using GIS and remote sensing techniques.
- Produced maps of structural mapping of faults, folds, and lineament identification and analysis.
- Produced maps of anomalous behavior, including Bouguer gravity anomalies.

EDUCATION

Certificate of Data analysis	M2i, France	Jun - Sept, 2025
MSc. Environmental hazards and risks management	Côte d'Azur university, France	2019-2021
MSc. Georesources	Tlemcen university, Algeria	2016 - 2018
BSc. Geoscience	Tlemcen university, Algeria	2012 - 2015
Advanced certificate level	Kibaha Secondary school , Tanzania	2009 - 2011
Ordinary level certificate	Mzumbe Secondary school, Tanzania	2005 - 2008

CERTIFICATES & AWARDS

- *Geospatial applications for risk management | UNOOSA and CSSTEAP*
- *Leadership and Conflict management and resolution | Excellencia training center*
- *Python for spatial analysis | Coursera*
- *Project management | LAMAABE*
- *French government scholarship |2019*
- *Algerian government scholarship | 2011*